

70019 Probabilistic Inference: Week 2 Exercises

Conjugacy, Gaussian Processes

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1 Background material

You should have a *firm* understanding of Mathematics for Machine Learning. We will be manipulating probabilities and expectations freely, as discussed in Mathematics for Machine Learning. If steps are difficult, I encourage you to discuss with your peers, GTAs, and myself on the course EdStem page.

- Basic probability: sample spaces, disjoint events (summation of probabilities), independent events (multiplication of probabilities). See [Walpole and Myers \(2012\)](#), Ch2 (Imperial Library, or search Google for reading options).
- Probability densities. See [Deisenroth et al. \(2020\)](#) §6.2.
- Sum, product & Bayes' rules. See [Deisenroth et al. \(2020\)](#) §6.3.
- Unconstrained continuous optimisation. See [Deisenroth et al. \(2020\)](#) §7.1.
- Linear algebra and matrix decompositions. See [Deisenroth et al. \(2020\)](#) chs 2–4.
- A familiarity with linear basis-function regression. See [Deisenroth et al. \(2020\)](#) ch 9.

2 Notation of probabilities

In this course we will use the notation for probabilities that is common in machine learning. The main advantage is that this notation is shorter, although it does leave certain things implicit.

- We generally denote outcomes of random variables without referring explicitly to the random variable itself. For example, when we refer to an outcome $\mathbf{x}$, we implicitly know there is a random variable that can take this value. We usually denote this as the capital, for example here X .
- If $\mathbf{x} \in \mathbb{N}^D \implies p(\mathbf{x})$ is a probability mass function, i.e. $p(\mathbf{x}) = P(X = \mathbf{x})$, i.e., the probability of the random variable X takes value $\mathbf{x}$.
- If $\mathbf{x} \in \mathbb{R}^D \implies p(\mathbf{x})$ is a density. So in this case, $P(a < \mathbf{X} < b) = \int_a^b p(\mathbf{x}) d\mathbf{x}$.
- If I want to be explicit about the random variable that we are evaluating the density/mass of, I will denote this using a subscript, e.g., $p_{X,Y}(\mathbf{x}, \mathbf{y}) = p_{X|Y}(\mathbf{x}|\mathbf{y})p_Y(\mathbf{y})$.
- Expectations are denoted as $\mathbb{E}_{p(\mathbf{x})}[f(\mathbf{x})] = \int p(\mathbf{x})f(\mathbf{x})d\mathbf{x}$, or with respect to a different density $\mathbb{E}_{q(\mathbf{x})}[h(\mathbf{x})] = \int q(\mathbf{x})h(\mathbf{x})d\mathbf{x}$.
- Often, densities and pmfs can be discussed in exactly the same way, if we think of the density of a discrete RV as a sum of delta functions. I.e. $p(\mathbf{x}) = \sum_o \delta(\mathbf{x} - \mathbf{x}_o)p_o$, where $\{\mathbf{x}_o\}$ are all the possible outcomes that X can take, and p_o are their corresponding probabilities. This allows us to write an expectation as an integral, regardless of whether the RV is continuous or discrete, because for discrete RVs we get:

$$\mathbb{E}_{p(\mathbf{x})}[f(\mathbf{x})] = \int p(\mathbf{x})f(\mathbf{x})d\mathbf{x} = \int \sum_o \delta(\mathbf{x} - \mathbf{x}_o)p_o f(\mathbf{x})d\mathbf{x} = \sum_o f(\mathbf{x}_o)p_o. \quad (1)$$

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- There are some reasons to cringe at this notation, but it is the norm in statistical learning, and it's actually very convenient in many ways.

3 Exercises

3.1 Conjugate Priors

Question 1 (Finding a Conjugate Prior). Consider a coin flipping experiment, where we observe N_1 heads and N_0 tails from a Bernoulli random variable, with an unknown probability of success θ , giving the likelihood for the data $D = \{N_1, N_0\}$:

$$p(D|\theta) = \theta^{N_1} \cdot (1 - \theta)^{N_0} = \ell(\theta). \quad (2)$$

Derive a conjugate prior for this likelihood.

Hints:

- Remember the form of the posterior: $p(\theta|D) \propto \ell(\theta)p(\theta)$.
- What form should $p(\theta)$ take, and what should its parameters be, so that after multiplying by $\ell(\theta)$, you get a posterior distribution with the same form, but different parameters.
- You don't need to find the normalising constant, but the answer will provide it. This requires knowing some integrals (for this question, the integral was also given in lecture). This is a good skill to have, but not the focus of the course.

Solution: We are asked to derive a conjugate prior $p(\theta)$ for a binomial likelihood

$$p(D|\theta) = \theta^{N_1} (1 - \theta)^{N_0}.$$

The posterior for a conjugate prior has the same form as the prior, but with different parameters, so we know that

$$p(\theta|D) = \frac{1}{Z} p(\theta) p(D|\theta) \quad (3)$$

$$= \frac{1}{Z} p(\theta) \theta^{N_1} (1 - \theta)^{N_0} \quad (4)$$

shares the form of $p(\theta)$. Z is a normalising constant which, as the question reminds us, we need not evaluate.

We can see that choosing a prior with similar form to the likelihood, we can simply collect the exponents of θ and $1 - \theta$ to get the corresponding parameters of the posterior. i.e.

$$p(\theta) = p(\theta; a, b) = \frac{1}{Z_\theta} \theta^a (1 - \theta)^b \quad (5)$$

$$p(\theta|D) = \frac{1}{Z'} \theta^{a+N_1} (1 - \theta)^{b+N_0} = p(\theta; a + N_1, b + N_0) \quad (6)$$

where $Z' = Z \cdot Z_\theta$ and a, b are chosen prior parameters (or hyperparameters). As we saw in lecture, this conjugate prior is in fact a *beta* distribution, which is more commonly written as

$$p(\theta; \alpha, \beta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)}. \quad (7)$$

$B(.,.)$ is known as the beta function. By comparing with the Bernoulli likelihood, we can see that $\alpha - 1$ and $\beta - 1$ act as pseudo-counts.

Question 2 (General Conjugate Prior for Exponential Family). For a conjugate family likelihood

$$\ell(\eta) = p(x|\eta) = h(x) \exp(\eta^T t(x) - A(\eta)) \quad (8)$$

- show that the conjugate prior is given by

$$p(\eta|\tau, n_0) = H(\tau, n_0) \exp(\tau^T \eta - n_0 A(\eta)), \quad (9)$$

- and put the conjugate prior into the natural parameterisation.

Solution: A prior is conjugate if the posterior is of the same family as the prior. The given likelihood is

$$\ell(\eta) = p(x|\eta) = h(x) \exp(\eta^T t(x) - A(\eta)), \quad (10)$$

and the prior is

$$p(\eta|\tau, n_0) = H(\tau, n_0) \exp(\tau^T \eta - n_0 A(\eta)). \quad (11)$$

We can find the posterior using

$$p(\eta|x, \tau, n_0) \propto p(\eta|\tau, n_0)p(x|\eta). \quad (12)$$

Computing this quantity is straightforward

$$p(\eta|\tau, n_0)p(x|\eta) = h(x)H(\tau, n_0) \exp(\eta^T t(x) - A(\eta) + \tau^T \eta - n_0 A(\eta)) \quad (13)$$

$$= H'(x, \tau, n_0) \exp((t(x) + \tau)^T \eta - (1 + n_0)A(\eta)), \quad (14)$$

where $H'(x, \tau, n_0) = h(x)H(\tau, n_0)$. As this is the same form as the prior, the prior is conjugate.

Question 3 (Exponential Family Conjugate Priors). Using exponential family relations, derive the conjugate priors for:

1. The Bernoulli likelihood.
2. A Gaussian likelihood with unknown precision $\nu = 1/\sigma^2$, but known mean μ .

Solution:

1. The likelihood of the probability of success θ for a Bernoulli-distributed random variable is given by

$$p(x|\theta) = \theta^x (1 - \theta)^{(1-x)}. \quad (15)$$

Putting this into the exponential family form given in the previous question

$$p(x|\theta) = \exp \log(\theta^x (1 - \theta)^{(1-x)}) \quad (16)$$

$$= \exp \left(x \log \frac{\theta}{1 - \theta} + \log(1 - \theta) \right) \quad (17)$$

we see that

$$h(x) = 1 \quad (18)$$

$$\eta = \log \frac{\theta}{1 - \theta} \quad (19)$$

$$t(x) = x \quad (20)$$

$$A(\eta) = -\log \left(1 - \frac{e^\eta}{1 + e^\eta} \right) = \log(1 + e^\eta). \quad (21)$$

Starting from the general exponential family conjugate prior we have shown in the previous question, we can find the Bernoulli conjugate prior

$$p(\eta|\tau, n_0) = H(\tau, n_0) \exp(\tau^T \eta - n_0 A(\eta)) \quad (22)$$

$$p(\theta|\tau, n_0) = H(\tau, n_0) \exp \left(\tau^T \log \frac{\theta}{1 - \theta} + n_0 \log(1 - \theta) \right) \quad (23)$$

$$= H(\tau, n_0) \theta^\tau (1 - \theta)^{n_0 - \tau}. \quad (24)$$

As discussed above, this is a beta distribution, with parameters $\alpha = \tau + 1$ and $\beta = n_0 - \tau + 1$. Its normalising constant is defined by the beta function: $H(\tau, n_0) = 1/B(\alpha, \beta)$.

2. The Gaussian likelihood, for a precision $\nu = 1/\sigma^2$ is

$$p(x|\mu, \nu) = \sqrt{\frac{\nu}{2\pi}} \exp\left(-\frac{\nu}{2}(x - \mu)^2\right) \quad (25)$$

$$= \sqrt{\frac{\nu}{2\pi}} \exp\left(-\frac{\nu}{2}(x^2 - 2x\mu + \mu^2)\right). \quad (26)$$

The parameters of the exponential family form are

$$h(x) = (2\pi)^{-\frac{1}{2}} \quad (27)$$

$$\eta = \nu \quad (28)$$

$$t(x) = -\frac{1}{2}(x^2 - 2x\mu) \quad (29)$$

$$A(\eta) = \frac{1}{2}(\nu\mu^2 - \log \nu). \quad (30)$$

Note that we have used the fact that μ is known and therefore does not appear in our natural parameters. Starting from the general exponential family conjugate prior, we can write

$$p(\eta|\tau, n_0) = H(\tau, n_0) \exp(\tau^T \eta - n_0 A(\eta)) \quad (31)$$

$$p(\nu|\mu, \tau, n_0) = H(\tau, n_0) \exp\left(\tau\nu - n_0 \frac{1}{2}(\nu\mu^2 - \log \nu)\right) \quad (32)$$

$$p(\nu|\mu, \tau, n_0) = H(\tau, n_0) \exp\left(\nu\left(\tau - \frac{n_0\mu^2}{2}\right)\right) \nu^{\frac{n_0}{2}}. \quad (33)$$

This is in fact called an *inverse gamma* distribution on σ^2 . Inverse gamma distributions are usually written in the form

$$p(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \left(\frac{1}{x}\right)^{\alpha+1} \exp(-\beta/x) \quad (34)$$

where $\Gamma(\alpha)$ is known as the gamma function.

Question 4 (Multivariate Gaussian Conjugacy). Consider a multivariate Gaussian likelihood for $p(\mathbf{y}|\mathbf{f}) = \mathcal{N}(\mathbf{y}; \mathbf{f}, \sigma^2 \mathbf{I}_N)$. Show that the prior $p(\mathbf{f}) = \mathcal{N}(\mathbf{f}; \mathbf{0}, \mathbf{K})$ is conjugate by placing both distributions into their natural exponential family form.

Solution: This question addresses multivariate Gaussian distributions: we will be setting all distributions in their natural form so we first tackle how to put in exponential family form the general multivariate Gaussian with mean $\boldsymbol{\mu}$ and covariance matrix Σ . For an N dimensional distribution with likelihood model $p(\mathbf{x}; \boldsymbol{\mu}, \Sigma)$, we start by rewriting

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi|\Sigma|)^{\frac{N}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \quad (35)$$

$$= \frac{1}{(2\pi)^{\frac{N}{2}}} \exp\left(-\frac{1}{2}\mathbf{x}^T \Sigma^{-1} \mathbf{x} + \mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu} - \frac{1}{2}\boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu} - \frac{N}{2} \log |\Sigma|\right). \quad (36)$$

To write this in the required exponential form, we have to rearrange the term in the exponential so that somehow we have a ‘sufficient statistic’ $t(x)$ term that is a function of the data dotted with vector η that is some function of the parameters $\boldsymbol{\mu}, \Sigma$. This is straightforward for the term $\mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu}$, and for the term quadratic in $\mathbf{x}$ we can use the following trick. We have

$$\mathbf{x}^T \Sigma^{-1} \mathbf{x} = \sum_{i,j} x_i \Sigma_{i,j}^{-1} x_j \quad (37)$$

$$= \sum_{i,j} x_i \Sigma_{i,j}^{-1} x_j \quad (38)$$

$$= \sum_{i,j} \Sigma_{i,j}^{-1} x_i x_j \quad (39)$$

$$= \overrightarrow{(\mathbf{x}\mathbf{x}^T)} \overrightarrow{(\Sigma^{-1})}, \quad (40)$$

where $\mathbf{x}\mathbf{x}^T$ is a matrix with element i, j equal to $x_i x_j$ and the notation $\overrightarrow{(\cdot)}$ denotes that we are flattening the relevant matrix into a vector (preserving the ordering so that all matrices are flattened in the same way). We have now written the quadratic term in the required form of a dot product (between two N^2 dimensional vectors) where one is a function of $\mathbf{x}$ and the other is in terms of the distribution parameters $\boldsymbol{\mu}, \Sigma$.

Substituting into our previous expression gives

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(-\frac{1}{2} \overrightarrow{\mathbf{x}\mathbf{x}^T}^T \Sigma^{-1} \overrightarrow{\mathbf{x}\mathbf{x}^T} + \mathbf{x}^T \Sigma^{-1} \boldsymbol{\mu} - \frac{1}{2} \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu} - \frac{N}{2} \log |\Sigma| \right) \quad (41)$$

$$= \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(\left[\begin{array}{c} \mathbf{x} \\ \overrightarrow{\mathbf{x}\mathbf{x}^T} \end{array} \right]^T \left[\begin{array}{c} \Sigma^{-1} \boldsymbol{\mu} \\ -\Sigma^{-1}/2 \end{array} \right] - \left(\frac{1}{2} \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu} + \frac{N}{2} \log |\Sigma| \right) \right) \quad (42)$$

and comparing with the standard form of the exponential family $p(\mathbf{x}|\eta) = h(\mathbf{x}) \exp(\eta^T t(\mathbf{x}) - A(\eta))$, we can now make the immediate identifications $h(\mathbf{x}) = (2\pi)^{-\frac{N}{2}}$, $t(\mathbf{x}) = \left[\mathbf{x} \quad \overrightarrow{\mathbf{x}\mathbf{x}^T} \right]^T$ and $\eta = \left[\Sigma^{-1} \boldsymbol{\mu} \quad -\Sigma^{-1}/2 \right]^T$. Finally, we identify $A(\eta) = \frac{1}{2} \boldsymbol{\mu}^T \Sigma^{-1} \boldsymbol{\mu} + \frac{N}{2} \log |\Sigma|$, which we could in principle rewrite as a function of the elements of η to determine $A(\eta)$ explicitly. For this question this implicit form of $A(\eta)$ will suffice.

Applying our general formula to the particular multivariate Gaussians of interest to the question we recover for the likelihood

$$p(\mathbf{y}|\mathbf{f}) = \mathcal{N}(\mathbf{y}; \mathbf{f}, \sigma^2 \mathbf{I}_N) = \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(\left[\begin{array}{c} \mathbf{y} \\ \overrightarrow{\mathbf{y}\mathbf{y}^T} \end{array} \right]^T \left[\begin{array}{c} \mathbf{f}/\sigma^2 \\ -\mathbf{I}_N/(2\sigma^2) \end{array} \right] - \frac{1}{2\sigma^2} \mathbf{f}^T \mathbf{f} - \frac{N}{2} \log \sigma^2 \right) \quad (43)$$

with $h_l(y) = (2\pi)^{-\frac{N}{2}}$, $\eta_l = \left[\mathbf{f}/\sigma^2 \quad -\mathbf{I}_N/(2\sigma^2) \right]^T$, $t_l(y) = \left[\mathbf{y} \quad \overrightarrow{\mathbf{y}\mathbf{y}^T} \right]^T$ and implicit $A_l(\eta_l) = \frac{1}{2\sigma^2} \mathbf{f}^T \mathbf{f} - \frac{N}{2} \log \sigma^2$.

Note that in this case the likelihood above could also be simplified as

$$p(\mathbf{y}|\mathbf{f}) = \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(-\frac{1}{2\sigma^2} \mathbf{y}^T \mathbf{y} + \frac{1}{\sigma^2} \mathbf{y}^T \mathbf{f} - \frac{1}{2\sigma^2} \mathbf{f}^T \mathbf{f} - \frac{N}{2} \log \sigma^2 \right) \quad (44)$$

$$= \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(\left[\begin{array}{c} \mathbf{y} \\ \mathbf{y}^T \mathbf{y} \end{array} \right]^T \left[\begin{array}{c} \mathbf{f}/\sigma^2 \\ -1/(2\sigma^2) \end{array} \right] - \frac{1}{2\sigma^2} \mathbf{f}^T \mathbf{f} - \frac{N}{2} \log \sigma^2 \right) \quad (45)$$

which would result in an alternative but equivalent parametrisation in exponential family form with lower dimensional η and $t(y)$ vectors.

Moving on to the prior, we recover

$$\mathcal{N}(\mathbf{f}; 0, \mathbf{K}) = \frac{1}{(2\pi)^{\frac{N}{2}}} \exp \left(\left[\begin{array}{c} \mathbf{f} \\ \overrightarrow{\mathbf{f}\mathbf{f}^T} \end{array} \right]^T \left[\begin{array}{c} \vec{0} \\ -\frac{1}{2} \mathbf{K}^{-1} \end{array} \right] - \frac{N}{2} \log |\mathbf{K}| \right) \quad (46)$$

with corresponding parametrisation $h_p(f) = (2\pi)^{-\frac{N}{2}}$, $\eta_p = \left[\vec{0} \quad -\frac{1}{2} \mathbf{K}^{-1} \right]^T$, $t_p(f) = \left[\mathbf{f} \quad \overrightarrow{\mathbf{f}\mathbf{f}^T} \right]^T$ and $A_p(\eta_p) = \frac{N}{2} \log |\mathbf{K}|$.

Now, to ascertain that the two distributions are conjugate, we look at the form of the posterior. By multiplying the two, we see it will be proportional to

$$p(f|y) \propto \exp \left(\left[\begin{array}{c} \mathbf{y} \\ \overrightarrow{\mathbf{y}\mathbf{y}^T} \end{array} \right]^T \left[\begin{array}{c} \mathbf{f}/\sigma^2 \\ -\mathbf{I}_N/(2\sigma^2) \end{array} \right] - \frac{1}{2\sigma^2} \mathbf{f}^T \mathbf{f} - \frac{N}{2} \log \sigma^2 + \left[\begin{array}{c} \mathbf{f} \\ \overrightarrow{\mathbf{f}\mathbf{f}^T} \end{array} \right]^T \left[\begin{array}{c} \vec{0} \\ -\frac{1}{2} \mathbf{K}^{-1} \end{array} \right] - \frac{N}{2} \log |\mathbf{K}| \right) \quad (47)$$

$$\propto \exp \left(\underbrace{\left[\begin{array}{c} \mathbf{f} \\ \overrightarrow{\mathbf{f}\mathbf{f}^T} \end{array} \right]^T}_{t(f)} \underbrace{\left[\begin{array}{c} \mathbf{y}/\sigma^2 \\ -\frac{1}{2} \left(\mathbf{K}^{-1} + \mathbf{I}_N/\sigma^2 \right) \end{array} \right]}_{\eta} - \underbrace{\left(\mathbf{y}^T \mathbf{K}^{-1} \mathbf{y} + \frac{N}{2} (\log \sigma^2 + \log |\mathbf{K}|) \right)}_{A(\eta)} \right) \quad (48)$$

and we have the posterior in exponential family form (so the two distributions are conjugate), with the normalisation constant h that could be found by normalising the expression with respect to $\mathbf{f}$.

3.2 From Linear Models to GPs

Question 5 (Complete certainty). Show that for a Bayesian Linear Regression model with M basis functions, the variance of the posterior becomes zero when you observe M data points with zero observation noise.

Solution: We assume the following model

$$p(\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{I}_M), \quad (49)$$

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \mathcal{N}(\boldsymbol{\Phi}(\mathbf{X})\boldsymbol{\theta}, \sigma^2 \mathbf{I}_N), \quad (50)$$

where $\boldsymbol{\Phi}(\mathbf{X})$ is a $N \times M$ matrix and M is the number of basis functions. The posterior on $\boldsymbol{\theta}$ will be Gaussian as

$$p(\boldsymbol{\theta}|\mathbf{y}, \mathbf{X}) \propto p(\mathbf{y}|\boldsymbol{\theta}, \mathbf{X})p(\boldsymbol{\theta}), \quad (51)$$

and both the likelihood and prior are Gaussian, and the likelihood is a linear function of the parameter. Looking at just the variance (you should be able to derive this on your own), we get

$$\Sigma_{\boldsymbol{\theta}|\mathbf{y}, \mathbf{X}} = \mathbf{I}_M - \boldsymbol{\Phi}(\mathbf{X})^T [\boldsymbol{\Phi}(\mathbf{X})\boldsymbol{\Phi}(\mathbf{X})^T + \sigma^2 \mathbf{I}_N]^{-1} \boldsymbol{\Phi}(\mathbf{X}). \quad (52)$$

If we have M distinct datapoints, $\boldsymbol{\Phi}(\mathbf{X})$ is an $M \times M$ matrix and is full rank and hence invertible. We take $\sigma^2 \rightarrow 0$ as that is the likelihood noise. Then,

$$\Sigma_{\boldsymbol{\theta}|\mathbf{y}, \mathbf{X}} = \mathbf{I}_M - \boldsymbol{\Phi}(\mathbf{X})^T (\boldsymbol{\Phi}(\mathbf{X})\boldsymbol{\Phi}(\mathbf{X})^T)^{-1} \boldsymbol{\Phi}(\mathbf{X})^{-1} \boldsymbol{\Phi}(\mathbf{X}). \quad (53)$$

$$= \mathbf{0} \quad (54)$$

Question 6 (Gaussian Processes). Let's consider the joint distribution in the GP setting, i.e., one that factorises as

$$p(\mathbf{y}, \mathbf{y}^*, \mathbf{f}, \mathbf{f}^* | X, X^*, \boldsymbol{\theta}) = \left[\prod_{n=1}^N p(y_n | f_n, \mathbf{x}_n) \right] \left[\prod_{t=1}^{N^*} p(y_t^* | f_t^*, \mathbf{x}_t^*) \right] p(\mathbf{f}, \mathbf{f}^* | X, X^*), \quad (55)$$

with the additional knowledge that

$$p(\mathbf{f} | X, X^*) = p(\mathbf{f} | X), \quad \text{and} \quad p(\mathbf{f}^* | X, X^*) = p(\mathbf{f}^* | X^*), \quad (56)$$

and the convention that $X = \{\mathbf{x}_n\}_{n=1}^N$, $\mathbf{x}_n \in \mathbb{R}^D$, and $X \in \mathbb{R}^{N \times D}$.

- Find $p(\mathbf{f} | \mathbf{y}, X)$ in terms of tractable densities.
- Find $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f} | \mathbf{y}, X, X^*)$ in terms of tractable densities.
- Given that you know how to find $p(\mathbf{f} | \mathbf{y}, X)$ in terms of tractable densities, find $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f} | \mathbf{y}, X, X^*)$ in terms of tractable densities and $p(\mathbf{f} | \mathbf{y}, X)$.
- Is $p(\mathbf{f} | \mathbf{y}, X) = p(\mathbf{f} | \mathbf{y}, X, X^*)$?
- Based on the joint density given, can you find $p(\mathbf{f})$ or $p(\mathbf{y})$? What additional information do you need?

Solution:

- (Find $p(\mathbf{f} | \mathbf{y}, X)$): Using Bayes rule, we know that

$$p(\mathbf{f} | \mathbf{y}, X) \stackrel{\text{AT}}{=} \frac{p(\mathbf{y} | \mathbf{f}, X) p(\mathbf{f} | X)}{p(\mathbf{y} | X)}. \quad (57)$$

$$(58)$$

We can now write the numerator in terms of tractable densities:

$$p(\mathbf{y} | \mathbf{f}, X) = \prod_{n=1}^N p(y_n | f_n, \mathbf{x}_n), \quad \text{Due to iid noise assumption.} \quad (59)$$

$$p(\mathbf{f} | X) = \int p(\mathbf{f}, \mathbf{f}^* | X, X^*) d\mathbf{f}^*. \quad \text{Due to marginalisation and } p(\mathbf{f} | X, X^*) = p(\mathbf{f} | X). \quad (60)$$

Now, looking at the denominator

$$p(\mathbf{y}|X) = \int p(\mathbf{y}, \mathbf{f}|X) d\mathbf{f}, \quad \text{Due to marginalisation.} \quad (61)$$

$$p(\mathbf{y}, \mathbf{f}|X) = p(\mathbf{y}|\mathbf{f}, X)p(\mathbf{f}|X), \quad \text{Always true due to chain rule.} \quad (62)$$

where we have already found $p(\mathbf{f}|X)$.

- (Find $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f}|\mathbf{y}, X, X^*)$): For $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f}|\mathbf{y}, X, X^*)$, we will apply the chain rule for probabilities

$$p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f}|\mathbf{y}, X, X^*) = p(\mathbf{y}^*|\mathbf{f}^*, \mathbf{f}, \mathbf{y}, X, X^*)p(\mathbf{f}, \mathbf{f}^*|\mathbf{y}, X, X^*). \quad \text{Always true.} \quad (63)$$

We can now write the numerator in terms of tractable densities:

$$p(\mathbf{y}^*|\mathbf{f}^*, \mathbf{f}, \mathbf{y}, X, X^*) = p(\mathbf{y}^*|\mathbf{f}^*, X^*), \quad \text{Model assumptions.} \quad (64)$$

$$p(\mathbf{f}, \mathbf{f}^*|\mathbf{y}, X, X^*) = \frac{p(\mathbf{y}|\mathbf{f}, \mathbf{f}^*, X, X^*)p(\mathbf{f}, \mathbf{f}^*|X, X^*)}{p(\mathbf{y}|X, X^*)} \quad \text{Always true.} \quad (65)$$

$$= \frac{p(\mathbf{y}|\mathbf{f}, X)p(\mathbf{f}, \mathbf{f}^*|X, X^*)}{p(\mathbf{y}|X, X^*)}. \quad \text{Model assumptions.} \quad (66)$$

The simplification of terms due to model assumptions are due to the factorisation of the probability of everything. The denominator can be expanded similarly to the first part of the question,

$$p(\mathbf{y}|X, X^*) = \int p(\mathbf{y}|\mathbf{f}, X, X^*)p(\mathbf{f}|X, X^*)d\mathbf{f} \quad (67)$$

$$= \int p(\mathbf{y}|\mathbf{f}, X)p(\mathbf{f}|X)d\mathbf{f} \quad \text{Model assumptions.} \quad (68)$$

$$= p(\mathbf{y}|X). \quad (69)$$

- (Find $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f}|\mathbf{y}, X, X^*)$ in terms of tractable densities and $p(\mathbf{f}|\mathbf{y}, X)$): We have already found $p(\mathbf{y}^*, \mathbf{f}^*, \mathbf{f}|\mathbf{y}, X, X^*)$. However looking at the term

$$p(\mathbf{f}, \mathbf{f}^*|\mathbf{y}, X, X^*) = \frac{p(\mathbf{y}|\mathbf{f}, X)p(\mathbf{f}, \mathbf{f}^*|X, X^*)}{p(\mathbf{y}|X)}, \quad (70)$$

we can write the last term in the numerator as

$$p(\mathbf{f}, \mathbf{f}^*|X, X^*) = p(\mathbf{f}^*|\mathbf{f}, X, X^*)p(\mathbf{f}|X), \quad \text{Chain rule.} \quad (71)$$

$$p(\mathbf{f}^*|\mathbf{f}, X, X^*) = \frac{p(\mathbf{f}^*, \mathbf{f}|X, X^*)}{p(\mathbf{f}|X)}. \quad \text{Bayes rule.} \quad (72)$$

We can now see that

$$p(\mathbf{f}, \mathbf{f}^*|\mathbf{y}, X, X^*) = p(\mathbf{f}|\mathbf{y}, X)p(\mathbf{f}^*|\mathbf{f}, X, X^*). \quad (73)$$

The advantage here is that if we have already computed $p(\mathbf{f}|\mathbf{y}, X)$, we much more easily compute $p(\mathbf{f}, \mathbf{f}^*|\mathbf{y}, X, X^*)$.

3.3 Gaussian processes

Question 7 (Distributions over functions). A Gaussian process is defined as a “collection of a possibly infinite number of random variables, any finite number of which is Gaussian distributed”. When using a Gaussian process as a distribution over functions, how do we use the random variables in this collection?

Solution: For an alternative explanation, see [Rasmussen and Williams \(2006, p. 13\)](#). A scalar function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ can be seen as a collection of scalar output variables indexed by some input $\mathbf{x} \in \mathbb{R}^D$. In this case where inputs lie in the vector space $\mathbb{R}^D$, the number of variables in the collection is infinite¹, since there are an infinite number of inputs.² We can consider finite subsets of the output variables that correspond to finite subsets of inputs. I.e. for a set of N inputs

$$X = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}, \quad (74)$$

¹Or more specifically, *uncountably* infinite, but for the purpose of this course this is a detail of minor importance.

²In the course, we will consider only inputs in $\mathbb{R}^D$. However you can have functions on many different types of inputs, leading to different sizes of the set of output variables. For example, you could have a function from a finite set: $\{0, 1, 2, 3, 4\} \rightarrow \mathbb{R}$.

which we can also denote as a matrix $X \in \mathbb{R}^{N \times D}$, we have the corresponding output set

$$f(X) = \{f(\mathbf{x}_1), \dots, f(\mathbf{x}_N)\}, \quad (75)$$

which we alternatively denote as a vector $f(X) \in \mathbb{R}^N$. If the entire function $f(\cdot)$ is a random variable with some distribution, this will also mean that the function outputs $f(X)$ will be a random variable with some distribution.

Going back to the abstract definition of a Gaussian process, we see that its definition as a collection of random variables matches what we found must hold for a function-valued random variable. The collection of random variables that form the Gaussian process can be identified with the random variables of the function outputs at specified inputs. The next part of the definition specifies that a finite subset of the random variables in the Gaussian process must be Gaussian distributed, i.e. for some arbitrary set of inputs X , the corresponding set of outputs $f(X)$ is Gaussian distributed

$$f(\cdot) \sim \mathcal{GP} \implies p(f(X)) = \mathcal{N}(f(X); \boldsymbol{\mu}_{f(X)}, \boldsymbol{\Sigma}_{f(X)}) , \quad (76)$$

with some mean $\boldsymbol{\mu}_{f(X)} \in \mathbb{R}^N$ and covariance matrix $\boldsymbol{\Sigma}_{f(X)} \in \mathbb{R}^{N \times N}$. The value of X determines *which* random variables in the collection of the GP we are considering, and therefore must also determine the value of $\boldsymbol{\mu}_{f(X)}$ and $\boldsymbol{\Sigma}_{f(X)}$.

So, long story short: When we use a Gaussian process as a distribution over functions, the random variables in the collection that makes up the GP are used as the *outputs of the function at some input locations*.

Question 8 (Gaussian Process shapes). Consider the Gaussian process notation

$$p(\mathbf{y}, \mathbf{y}^*) = \mathcal{N}\left(\begin{bmatrix} \mathbf{y} \\ \mathbf{y}^* \end{bmatrix}; \mathbf{0}, \begin{bmatrix} k(X, X) & k(X, X^*) \\ k(X^*, X) & k(X^*, X^*) \end{bmatrix}\right), \quad (77)$$

Given that $\mathbf{y} \in \mathbb{R}^N$ and $\mathbf{y}^* \in \mathbb{R}^{N^*}$, what are the shapes of $k(X, X)$, $k(X, X^*)$, $k(X^*, X)$, $k(X^*, X^*)$

Solution: The shapes can be easily inferred from the multivariate Gaussian, e.g., by considering marginalisation. In particular, $k(X, X) \in \mathbb{R}^{N \times N}$, $k(X, X^*) \in \mathbb{R}^{N \times N^*}$, $k(X^*, X) \in \mathbb{R}^{N^* \times N}$, $k(X^*, X^*) \in \mathbb{R}^{N^* \times N^*}$.

Question 9 (Determining a GP*). The properties of a Gaussian process are fully determined by its mean and covariance function. T/F?

Solution: True

Question 10 (GP prior density*). Write down the probability density of the function values at input locations $X = \{\mathbf{x}_1, \dots, \mathbf{x}_5\}$ of a GP with mean function $\mu(\mathbf{x})$ and covariance function $k(\mathbf{x}, \mathbf{x}')$.

Solution: The mean function $\mu : \mathbb{R}^D \rightarrow \mathbb{R}$ and covariance function³ $k : \mathbb{R}^D \times \mathbb{R}^D \rightarrow \mathbb{R}$ determine the mean vector and covariance matrix of the distribution over function values for some input. For example, for the inputs X , we have

$$p(f(X)) = \mathcal{N}(f(X); \boldsymbol{\mu}_{f(X)}, \boldsymbol{\Sigma}_{f(X)}) , \quad (78)$$

$$\text{with } [\boldsymbol{\mu}_{f(X)}]_n = \mu(\mathbf{x}_n), \quad \text{and } [\boldsymbol{\Sigma}_{f(X)}]_{nn'} = k(\mathbf{x}_n, \mathbf{x}_{n'}), \quad (79)$$

where $[\dots]_n$ denotes indexing a vector or matrix. We will often denote the covariance matrix of a common set of inputs (e.g. training inputs X) as $\mathbf{K}$. Alternatively, we can denote the matrix resulting from evaluating the covariance function at two sets of inputs $X \in \mathbb{R}^{N \times D}$, $X^* \in \mathbb{R}^{N^* \times D}$ as

$$K(X, X^*) \in \mathbb{R}^{N \times N^*} . \quad (80)$$

Question 11 (GP posterior*). To do regression over an unknown function $f(\cdot)$, we use a GP prior $f(\cdot) \sim \mathcal{GP}(0, k(\mathbf{x}, \mathbf{x}'))$. We observe N noisy observations $\mathbf{y} = \{y_1, \dots, y_N\}$ which are i.i.d. distributed according to the likelihood $p(y_n | f(\mathbf{x}_n), \mathbf{x}_n) = \mathcal{N}(y_n; f(\mathbf{x}_n), \sigma^2)$. What is the posterior over the function values at an arbitrary set of input locations X^* .

³The covariance function is often referred to as the *kernel*.

Solution: In inference problems, the goal is to go from a joint probability distribution over all relevant variables (which comes from the model specification) to a conditional distribution given observations (the posterior). The posterior is obtained by manipulating the joint using the rules of probability. Often Bayes' rule is enough, but sometimes other applications of the product or sum rules are necessary (MacKay, 2003, p. 24). I will go through this derivation multiple ways, and in quite a lot of detail. I will pay special attention to the manipulations of the probability distributions on the way, as it is important to be familiar with how to do this. If you struggle with this, I would recommend looking over chapters 1–3 of MacKay (2003), and perhaps doing a few exercises.

3.3.1 Understanding and manipulating the joint

The joint distribution is usually specified through a set of conditional probability distributions. For Gaussian process regression, the problem specification gives:

1. The *prior* distribution over the function values at any inputs, through the definition of the prior over functions being a GP. For a set of inputs X , we can refer to the prior density over corresponding function values as $p(f(X))$.
2. The conditional distribution of any observation y_n at an input location $\mathbf{x}_n$, conditioned on full knowledge of the underlying function value. This distribution is known as the *likelihood*, and is written as $p(y_n | f(\mathbf{x}_n), \mathbf{x}_n)$ in this case, and has the density $\mathcal{N}(y_n; f(\mathbf{x}_n), \sigma^2)$.

We need to consider the function values at both the training points $f(X)$ and testing points $f(X^*)$, in addition to the outputs $\mathbf{y}$. This gives us the joint distribution $p(f(X), f(X^*), \mathbf{y} | X)$. We can factorise the joint along the conditional independence relationships that the model was specified with:

$$p(f(X), f(X^*), \mathbf{y} | X) = p(f(X), f(X^*))p(\mathbf{y} | f(X), X) \quad (81)$$

$$= p(f(X), f(X^*)) \prod_{n=1}^N p(y_n | f(\mathbf{x}_n), \mathbf{x}_n). \quad (82)$$

We can also go one step further and factorise the prior using ????:

$$p(f(X), f(X^*)) = \mathcal{N} \left(\begin{bmatrix} f(X) \\ f(X^*) \end{bmatrix}; 0, \begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{ff^*} \\ \mathbf{K}_{f^*f}^T & \mathbf{K}_{f^*f^*} \end{bmatrix} \right) \quad (83)$$

$$\implies p(f(X^*) | f(X)) = \mathcal{N} \left(f(X^*); \mathbf{K}_{f^*f}^T \mathbf{K}_{ff}^{-1} f(X), \mathbf{K}_{f^*f^*} - \mathbf{K}_{f^*f}^T \mathbf{K}_{ff}^{-1} \mathbf{K}_{ff^*} \right), \quad (84)$$

where $\mathbf{K}_{ff} = k(X, X)$, $\mathbf{K}_{ff^*} = k(X, X^*)$, and $\mathbf{K}_{f^*f^*} = k(X^*, X^*)$. This gives an overall factorised joint of

$$p(f(X), f(X^*), \mathbf{y} | X) = p(f(X))p(f(X^*) | p(f(X))) \prod_{n=1}^N p(y_n | f(\mathbf{x}_n), \mathbf{x}_n) \quad (85)$$

This conditional independence relationship can be represented as a graphical model, which often helps in reasoning about the model (fig. 1)



Figure 1: Graphical model representations of Gaussian process regression, with conditional independence of $\mathbf{y}$ represented explicitly on the right.

We are tasked with finding the posterior over function values at an arbitrary set of input locations X^* , given that we observe a training set consisting of N inputs and corresponding outputs $\{\mathbf{x}_n, y_n\}_{n=1}^N$. We arrange the inputs into a matrix $X \in \mathbb{R}^{N \times D}$, and the outputs into a vector $\mathbf{y} \in \mathbb{R}^N$.

3.3.2 Method 1: Starting with the appropriate joint

One representation of Bayes' rule that we can apply is the joint distribution divided by the conditional. In our case this becomes

$$p(f(X^*) | \mathbf{y}, X) = \frac{p(\mathbf{y}, f(X^*) | X)}{p(\mathbf{y} | X)}. \quad (86)$$

However, the model specification did not give us the density of $p(\mathbf{y}, f(X^*) | X)$, so we will have to find it ourselves. We can find it by marginalising (or *integrating out*) the unused variable $f(X)$ from the joint using the sum rule:

$$p(\mathbf{y}, f(X^*)) = \int p(\mathbf{y}, f(X), f(X^*) | X) d f(X). \quad (87)$$

We know the joint is a Gaussian, as the conditionals are all Gaussians which only interact linearly. This makes the marginal Gaussian too. We could perform the integration by hand, but given the knowledge that $p(\mathbf{y}, f(X^*) | X)$ is Gaussian, we could also just compute its means and covariances. The means are all zero. The main useful insight into finding the covariances, is that the Gaussian likelihood can be rewritten as

$$\mathbf{y} = f(X) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_N). \quad (88)$$

This makes the covariances

$$\text{Cov}[\mathbf{y}, f(X^*)] = \mathbb{E}_{f, \epsilon} [(f(X) + \epsilon) f(X^*)^T] = \mathbb{E}_{f, \epsilon} [f(X) f(X^*)^T] + \mathbb{E}_{f, \epsilon} [\epsilon f(X^*)^T] \quad (89)$$

$$= k(X, X^*) = \mathbf{K}_{ff^*}, \quad (90)$$

$$\text{Cov}[\mathbf{y}, \mathbf{y}] = \mathbb{E}_{f, \epsilon} [(f(X) + \epsilon)(f(X) + \epsilon)^T] = k(X, X) + \sigma^2 \mathbf{I} = \mathbf{K}_{ff} + \sigma^2 \mathbf{I}, \quad (91)$$

$$\text{Cov}[f(X^*), f(X^*)] = k(X^*, X^*) = \mathbf{K}_{f^*f^*}. \quad (92)$$

The posterior can now be found by simple Gaussian conditioning, using ????:

$$p(\mathbf{y}, f(X^*) | X) = \mathcal{N} \left(\begin{bmatrix} \mathbf{y} \\ f(X^*) \end{bmatrix}; 0, \begin{bmatrix} \mathbf{K}_{ff} + \sigma^2 \mathbf{I}_N & \mathbf{K}_{ff^*} \\ \mathbf{K}_{ff^*}^T & \mathbf{K}_{f^*f^*} \end{bmatrix} \right) \quad (93)$$

$$\implies p(f(X^*) | \mathbf{y}, X) = \mathcal{N}(f(X^*); \mathbf{K}_{ff^*}^T (\mathbf{K}_{ff} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}, \mathbf{K}_{f^*f^*} - \mathbf{K}_{ff^*}^T (\mathbf{K}_{ff} + \sigma^2 \mathbf{I})^{-1} \mathbf{K}_{ff^*}). \quad (94)$$

This completes the exercise.

3.3.3 Method 2: Finding a larger posterior then marginalising

Alternatively, we may want to apply Bayes' rule directly. However, our joint also contains $f(X)$, which we weren't interested in representing in our posterior. We can also solve the problem by applying Bayes' rule to find the posterior $p(f(X), f(X^*) | \mathbf{y}, X)$, and then marginalising out $f(X)$. We start by applying Bayes' rule to the usual prior-likelihood factorisation of the joint:

$$p(f(X), f(X^*) | \mathbf{y}, X) = \frac{p(\mathbf{y} | f(X), X) p(f(X), f(X^*))}{p(\mathbf{y} | X)}. \quad (95)$$

The first thing we can notice is that using the factorisation of the prior from eq. (84), we can actually write the posterior over both training and testing points in terms of the posterior over only the training points:

$$p(f(X), f(X^*) | \mathbf{y}, X) = \frac{p(\mathbf{y} | f(X), X) p(f(X))}{p(\mathbf{y} | X)} p(f(X^*) | f(X)) \quad (96)$$

$$= p(f(X) | \mathbf{y}) p(f(X^*) | f(X)) \quad (97)$$

We then marginalise out $f(X)$ to get the posterior we are interested in:

$$p(f(X^*) | \mathbf{y}) = \int p(f(X) | \mathbf{y}) p(f(X^*) | f(X)) d f(X). \quad (98)$$

Performing these computations requires exactly the same procedure of forming the joint and performing Gaussian conditioning. Once the joint $p(f(X), f(X^*) | \mathbf{y})$ is obtained, marginalisation can be performed simply by dropping the rows and columns related to $f(X)$ from the covariance. Overall the process is a bit more laborious than the other method.⁴

Question 12 (Marginalisation). Why do we not need to consider any points other than the training and testing points when doing regression over an entire function?

Solution: The Gaussian process prior defines a joint distribution over all finite subsets of the infinite set of function values. This raises the question of *which* finite subset we need to represent in order to do inference. We know that we at least need to represent $f(X)$ for the training data X , since these function values are involved in the likelihood. If we want to make predictions over some new subset of input points X^* , these need to be involved in inference as well. The question remains, do we need to represent any other points?

We only need to represent variables that influence the inference over variables we are interested in. To consider whether we need to represent any other function values, we can include them in our posterior analysis, integrate them out, and see whether they influence our beliefs in what we are interested in. We can do this by repeating the derivation in section 3.3.3, but with extra function values $f(\tilde{X})$ included.

$$\begin{aligned} p(f(X), f(X^*), f(\tilde{X}) | \mathbf{y}, X) &= \frac{p(\mathbf{y} | f(X), X) p(f(X))}{p(\mathbf{y} | X)} p(f(X^*) | f(X)) p(f(\tilde{X}) | f(X), f(X^*)) \quad (99) \\ &= p(f(X) | \mathbf{y}) p(f(X^*) | f(X)) p(f(\tilde{X}) | f(X), f(X^*)) \quad (100) \end{aligned}$$

By integrating out $f(\tilde{X})$ we see that the posterior given by this procedure is the same as the posterior which we obtained without considering the extra points $f(\tilde{X})$. Because this holds for *any* set of other points $\tilde{X}$, this shows that including any other set of points doesn't change the outcome of our inference procedure, and therefore we don't have to include them in the first place.

If you look at the derivation, you actually see that the fact that the prior over function values defines consistent distributions over function values. The distributions are consistent in the sense that if we consider the prior joint $p(f(X), f(X^*))$ and integrate out $f(X^*)$, then we get the same distribution as if we had considered the prior $p(f(X))$ directly. This is a direct consequence of a Gaussian process being a distribution over functions.

It is, of course, possible to define an infinite collection of probability densities that do not have this property. For example, defining the precision of a Gaussian using a covariance function. See [Rasmussen and Williams \(2006, §2.2\)](#) for another explanation.

Question 13 (Predicting observations*). How does the predictive distribution of a new *observation* at an input point $\mathbf{x}^*$ differ from the posterior over a function value $f(\mathbf{x}^*)$ (calculated in Q4)?

Solution: An observation y_n is modelled by the function value $f(\mathbf{x}_n)$ with some noise added. This question is asking for a posterior distribution over a different random variable than question 11 (i.e. y_n , rather than $f(\mathbf{x}_n)$). Once you have the posterior of $f(\mathbf{x}_n)$, you can use the usual manipulations of Gaussian distributions to find out that you just need to add $\sigma^2 \mathbf{I}_{N^*}$ to the covariance, where N^* is the number of test points. You could also apply Bayes' rule to the joint distribution

$$p(\mathbf{y}^* | \mathbf{y}, X, X^*) = \frac{p(\mathbf{y}, \mathbf{y}^* | X, X^*)}{p(\mathbf{y} | X)} \quad (101)$$

$$= \mathcal{N} \left(\begin{bmatrix} \mathbf{y} \\ \mathbf{y}^* \end{bmatrix}; 0, \begin{bmatrix} \mathbf{K}_{ff} + \sigma^2 \mathbf{I}_N & \mathbf{K}_{ff^*} \\ \mathbf{K}_{ff^*}^T & \mathbf{K}_{f^*f^*} + \sigma^2 \mathbf{I}_{N^*} \end{bmatrix} \right), \quad (102)$$

as in section 3.3.2.

Question 14 (Weights to covariances*). Say we parameterise multivariate linear functions as $f(\mathbf{x}) = \boldsymbol{\theta}_1^T \mathbf{x} + \theta_2$, and we define a prior over them by defining the priors over parameters as

$$p(\boldsymbol{\theta}_1) = \mathcal{N}(\boldsymbol{\theta}_1; 0, v_1 \mathbf{I}), \quad p(\theta_2) = \mathcal{N}(\theta_2; 0, v_2). \quad (103)$$

This actually defines a Gaussian process over $f(\cdot)$. Can you find its mean and covariance functions? Can you apply the same procedure to get the covariance function over functions parameterised as quadratics or higher-order polynomials?

⁴However, it does give us the nice insight that the only part of the posterior that *really* changes due to the data is $p(f(X) | \mathbf{y})$, and that any change in the posterior on $f(X^*)$ only occurs *through* the effect on $f(X)$.

Solution: In the question, a function is defined through a specific setting of the parameters. By making the parameters a random variable, we also make the function a random variable. By noting that we place Gaussian distributions on the parameters, and the relationship between the function value and parameters is linear, we see that the distribution over function values is also Gaussian. We can find the covariance function by simply finding the covariance between the function at two points. We first note that the mean function is zero by taking the mean.

$$k(\mathbf{x}, \mathbf{x}') = \mathbb{E}_{\boldsymbol{\theta}_1, \theta_2} [f(\mathbf{x})f(\mathbf{x}')] = \mathbb{E}_{\boldsymbol{\theta}_1, \theta_2} \left[(\mathbf{x}^T \boldsymbol{\theta}_1 + \theta_2)(\mathbf{x}'^T \boldsymbol{\theta}_1 + \theta_2) \right] \quad (104)$$

$$= \mathbf{x}^T \mathbf{x}' v_1 + v_2 \quad (105)$$

Question 15 (Sums of GPs*). We want to learn $f(\cdot)$, which can be decomposed as a sum of two functions:

$$f(\mathbf{x}) = f_1(\mathbf{x}) + f_2(\mathbf{x}). \quad (106)$$

If we know that $f_1 \perp f_2$, and that the covariance functions for $f_1(\cdot)$ and $f_2(\cdot)$ are $k_1(\cdot, \cdot)$ and $k_2(\cdot, \cdot)$ respectively, what will the prior over $f(\cdot)$ be? Specify the type of distribution and its parameters. If we make observations at $f(X)$ what is the posterior for $f_1(\cdot)$?

Solution: The prior over any set of points $f(X)$ is the sum of two independent random variables. Therefore, variances add.

$$f(X) = f_1(X) + f_2(X), \quad f_1(X) \sim \mathcal{N}(0, \mathbf{k}_1(X, X)), \quad f_2(X) \sim \mathcal{N}(0, k_2(X, X)), \quad (107)$$

$$p(f(X)) = \mathcal{N}(0, k_1(X, X) + k_2(X, X)) \quad (108)$$

Now, we want the posterior over $f_1(\cdot)$. We follow the same procedure as in question 11, by first finding the joint over all variables of interest, and then using the Gaussian conditioning formula.

$$\begin{aligned} \text{Cov}[f(X), f_1(X^*)] &= \mathbb{E}[(f_1(X) + f_2(X))f_1(X^*)^T] = \mathbb{E}[f_1(X)f_1(X^*)^T] + \mathbb{E}[f_1(X)f_2(X^*)^T] \\ &= k_1(X, X^*) \end{aligned} \quad (109)$$

$$p\left(\begin{bmatrix} f(X) \\ f_1(X^*) \end{bmatrix}\right) = \mathcal{N}\left(\begin{bmatrix} f(X) \\ f_1(X^*) \end{bmatrix}; \mathbf{0}, \begin{bmatrix} k_1(X, X) + k_2(X, X) & k_1(X, X^*) \\ k_1(X^*, X) & k_1(X^*, X^*) \end{bmatrix}\right) \quad (110)$$

$$\begin{aligned} \Rightarrow p(f_1(X^*) | f(X)) &= \mathcal{N}(f_1(X^*); k_1(X^*, X) [k_1(X, X) + k_2(X, X)]^{-1} f(X), \\ &\quad k_1(X^*, X^*) - k_1(X^*, X) [k_1(X, X) + k_2(X, X)]^{-1} k_1(X, X^*)) \end{aligned} \quad (111)$$

3.4 Model selection

Question 16 (Hyperparameter Conjugacy). Consider GP prior with a Squared Exponential kernel and uniform priors on its hyperparameters. Can you find the posterior on the hyperparameters $p(\boldsymbol{\theta}|\mathbf{y})$ in closed form?

Solution: Can you find the posterior on the hyperparameters $p(\boldsymbol{\theta}|\mathbf{y})$ in closed form? No, because the kernel (covariance) depends on l through some complicated non-linear function which we can't integrate over. Hence, we can't define a valid density in closed form that has the same functional form as a covariance matrix of a normal distribution with the non linearities imposed by the kernel.

When can we find a posterior in closed form? To find the posterior of the hyperparameters we can write, using Bayes rule, (leaving out dependence on $\mathbf{X}$ for this question)

$$p(\boldsymbol{\theta}|\mathbf{y}) = \frac{p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathbf{y})} \quad (112)$$

where $p(\mathbf{y}|\boldsymbol{\theta}) = \int p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\boldsymbol{\theta})d\boldsymbol{\theta}$ and $p(\mathbf{y}) = \int p(\mathbf{y}|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta}$. We know that $p(\mathbf{y}|\boldsymbol{\theta})$ is the marginal likelihood, and if we can find the posterior in closed form, we can compute the normaliser above $p(\mathbf{y})$. So, we ignore the normaliser for now. The given GP prior is

$$p(\mathbf{f}|\boldsymbol{\theta}) = \mathcal{N}(\mathbf{0}, \mathbf{K}_{SE}), \quad (113)$$

$$K_{SE}(x, x') = \lambda^2 \exp\left(-\frac{(x - x')^2}{2l^2}\right). \quad (114)$$

This will give us a marginal likelihood of

$$p(\mathbf{y}|\boldsymbol{\theta}) = (2\pi)^{-n/2} |\mathbf{K}_{SE} + \sigma_n^2 \mathbf{I}|^{-1/2} \exp \left(-\frac{1}{2} \mathbf{y}^T (\mathbf{K}_{SE} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \right) \quad (115)$$

where we have assumed $p(\mathbf{y}|\mathbf{f}) = \mathcal{N}(\mathbf{f}, \sigma_n^2 \mathbf{I})$, and $\mathbf{y} \in \mathbb{R}^n$. We can see that this is a Gaussian with covariance $\mathbf{K}_{SE} + \sigma_n^2 \mathbf{I}$. We know from conjugacy of Gaussian distributions that the conjugate prior for the mean of a multivariate Gaussian is a multivariate Gaussian. The conjugate prior for a covariance matrix $\boldsymbol{\Sigma}$ is an inverse Wishart distribution (multivariate form of the inverse Gamma)

$$p(\boldsymbol{\Sigma}|\boldsymbol{\Psi}, \nu) = \frac{|\boldsymbol{\Psi}|^{\nu/2}}{2^{\nu d/2} \Gamma_d(\frac{\nu}{2})} |\boldsymbol{\Sigma}|^{-(\nu+d+1)/2} \exp \left(-\frac{1}{2} \text{Tr}(\boldsymbol{\Psi} \boldsymbol{\Sigma}^{-1}) \right), \quad (116)$$

where $\boldsymbol{\Psi} \in \mathbb{R}^{d \times d}$ is a scale matrix and Γ_d is a multivariate Gamma function. It is easy to see why this is conjugate to the covariance function, it has a term that matches the inverse square root of the determinant $|\boldsymbol{\Sigma}|^{-(\nu+d+1)/2}$, and a term that matches the inverse in the exponential $\exp(-\frac{1}{2} \text{Tr}(\boldsymbol{\Psi} \boldsymbol{\Sigma}^{-1}))$. Thus, if we put an inverse Wishart prior on the covariance of the marginal likelihood, $\boldsymbol{\Sigma} = \mathbf{K}_{SE} + \sigma_n^2 \mathbf{I}$, then we could derive the posterior in closed form. However, if we wish to put a prior on the hyperparameters of $\mathbf{K}_{SE}$, λ and l , the conjugate prior needs to take into account the form of the kernel $\mathbf{K}_{SE}$. For λ , if we have $\sigma_n^2 = 0$, we can see that it essentially is a variance term, and the conjugate prior would be the inverse Gamma. For l , with $\sigma_n^2 = 0$, we would need a distribution that has a term with a determinant of the exponential of the inverse random variable, as well as a term that had an exponential of an exponential of the random variable. As there is no tractable distribution (that we know of) with these properties, we can't find the posterior for l in closed form.

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